

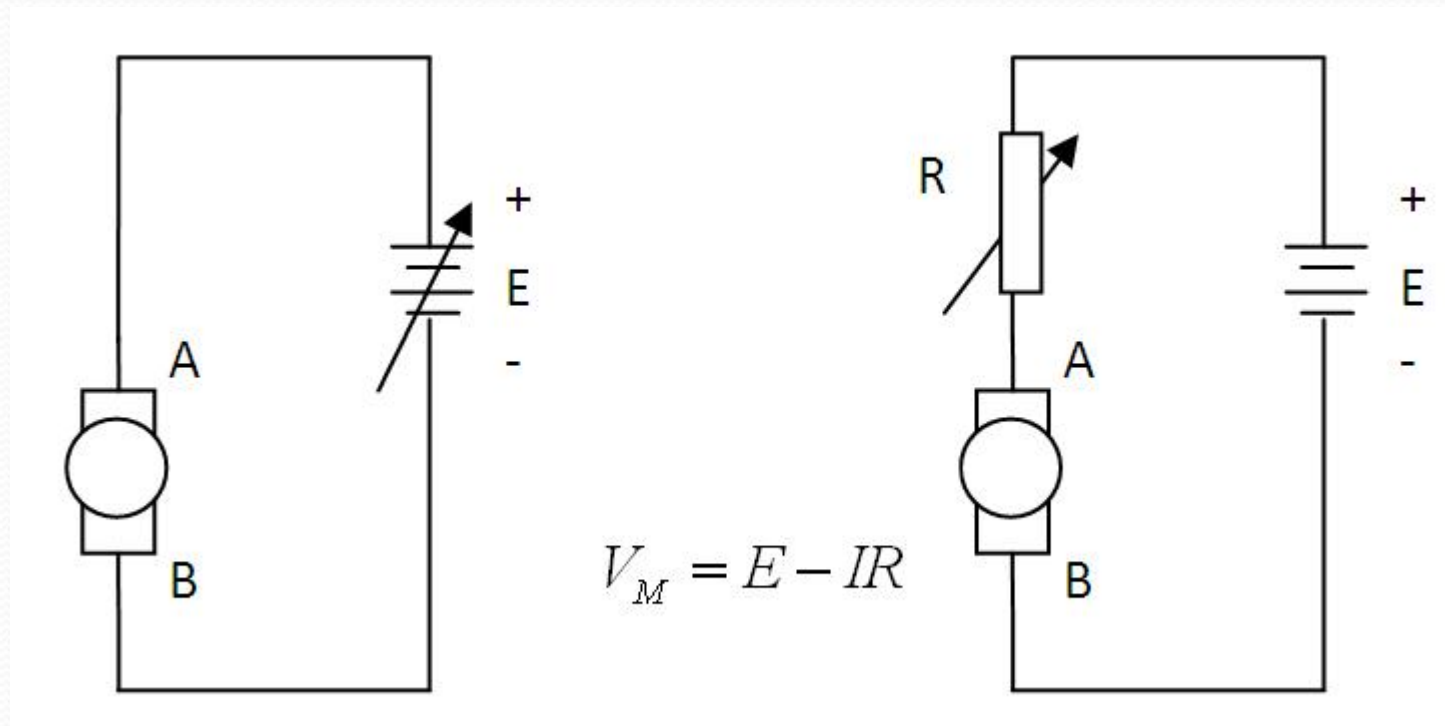
PWM电机调速控制

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□ 智能小车速度控制

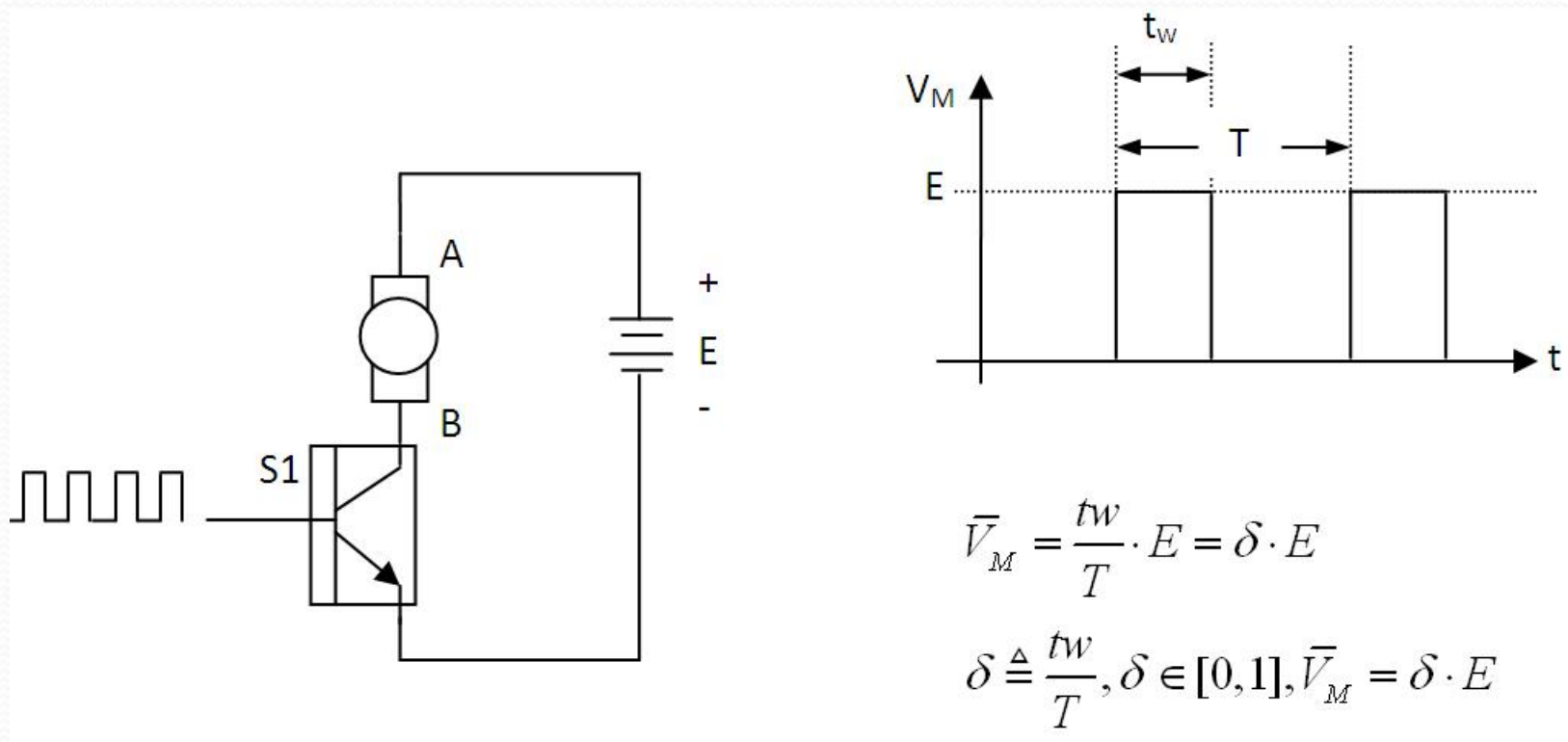
- 何时需要控制小车速度
- 小车速度控制的常用方法
- 调速过程的控制策略

■ 直接调压调速法原理



- 电机两端电压低时，扭矩小，可能无法启动
- 调压电阻产生能耗，产生高热

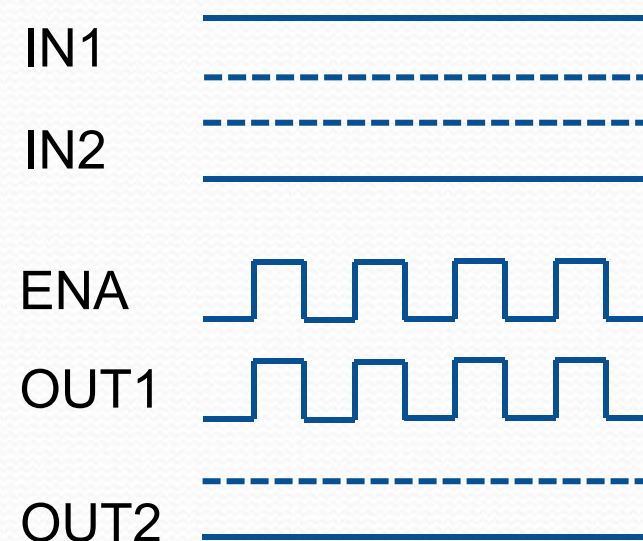
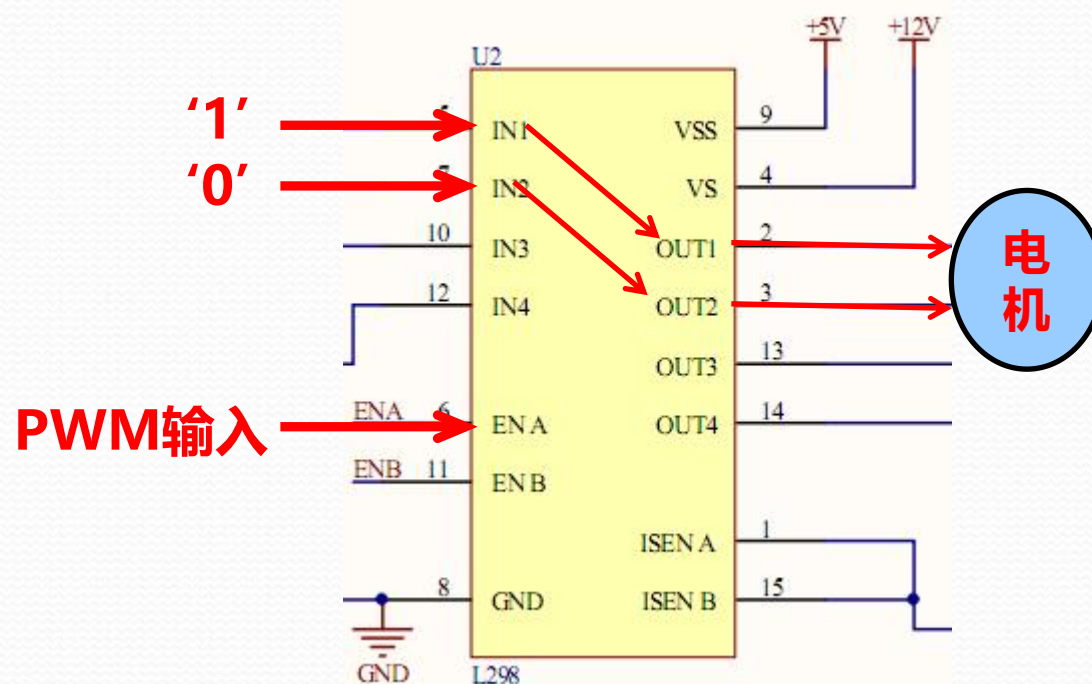
PWM调压调速法原理



- 利用间歇导通和电机运动惯性调速
- 电机两端电压始终等于 E ，扭矩大
- $S1$ 导通时， $V_{CE}=0$,控制功耗小

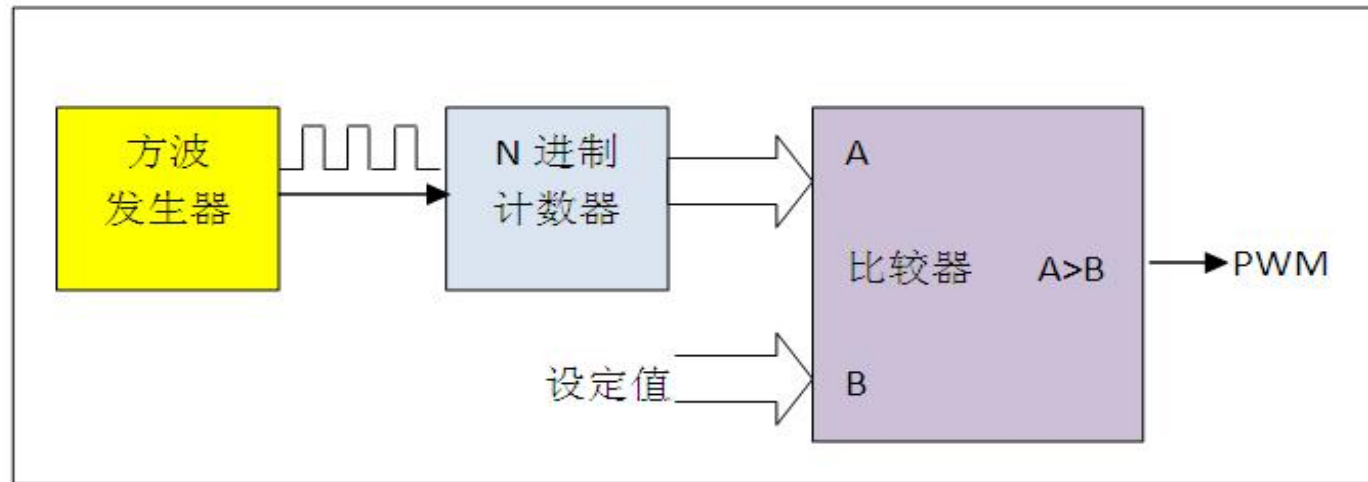
■ PWM调压调速法原理

• H桥的PWM调速示意图

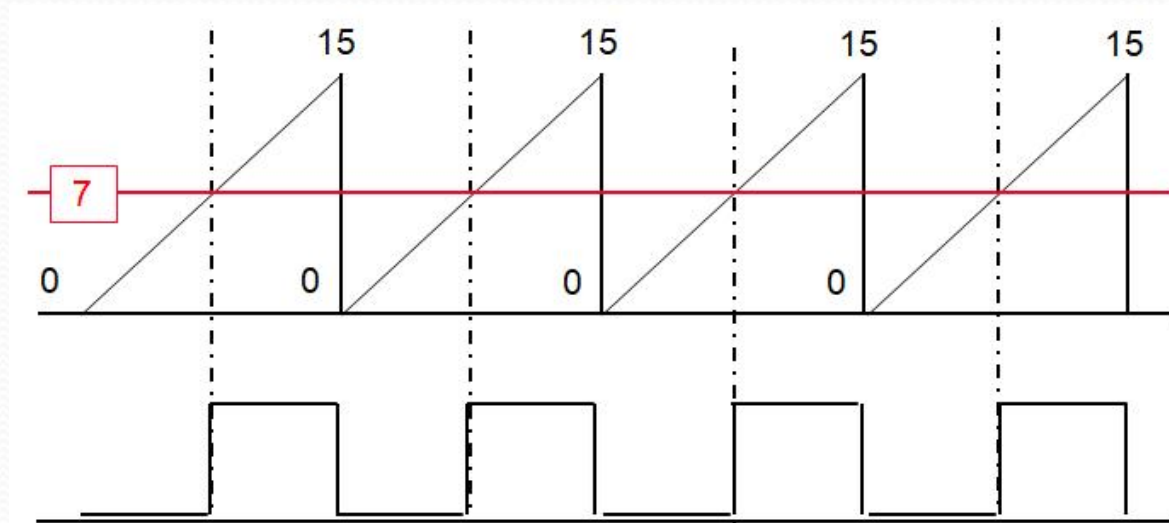


只适用于“启动-停止”模式切换，
不适用于“正转-反转”模式切换

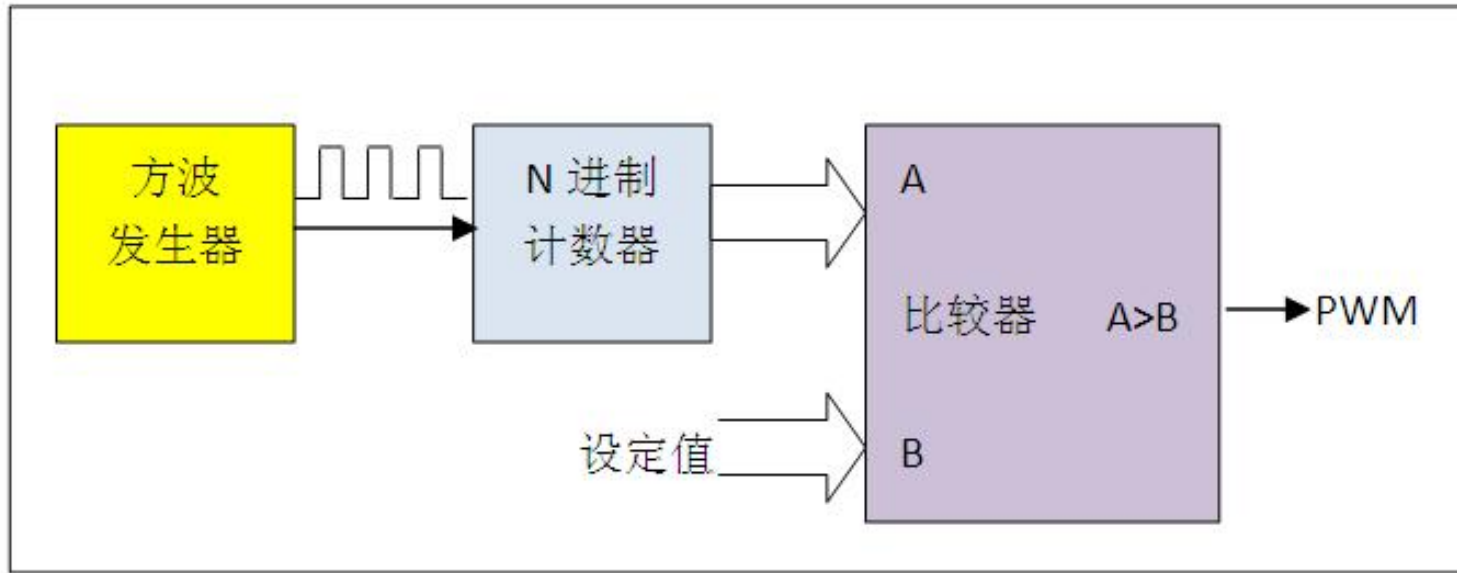
■ 数字式PWM电路



- 计数器输出产生锯齿波
- 比较器输出产生PWM波



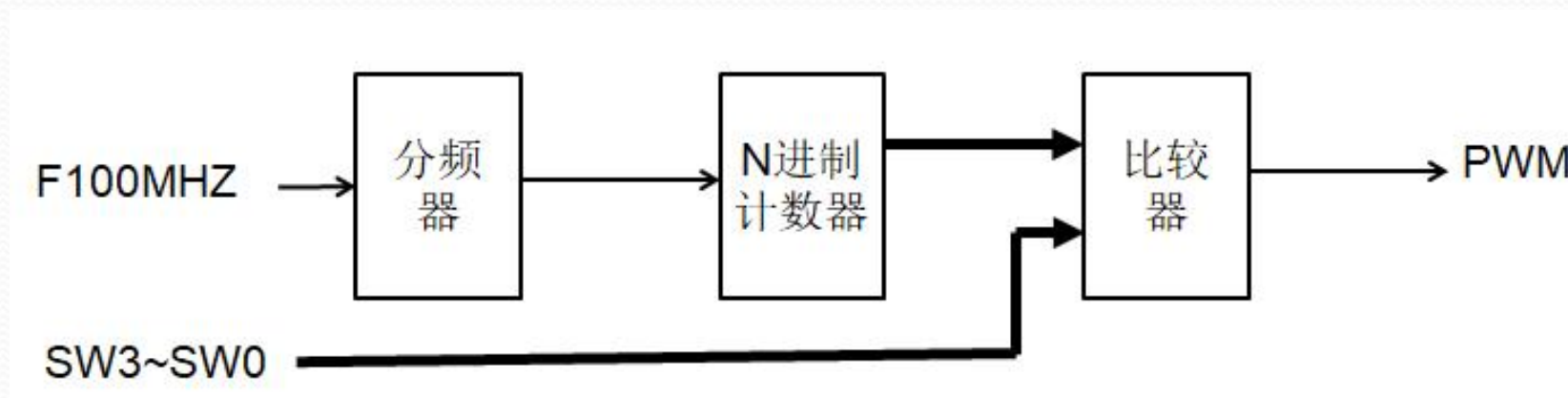
■ 数字式PWM电路的参数



- 若方波发生器的频率是 f
- PWM波的频率等于 f/N
- PWM波的占空比由比较设定值定，共N档

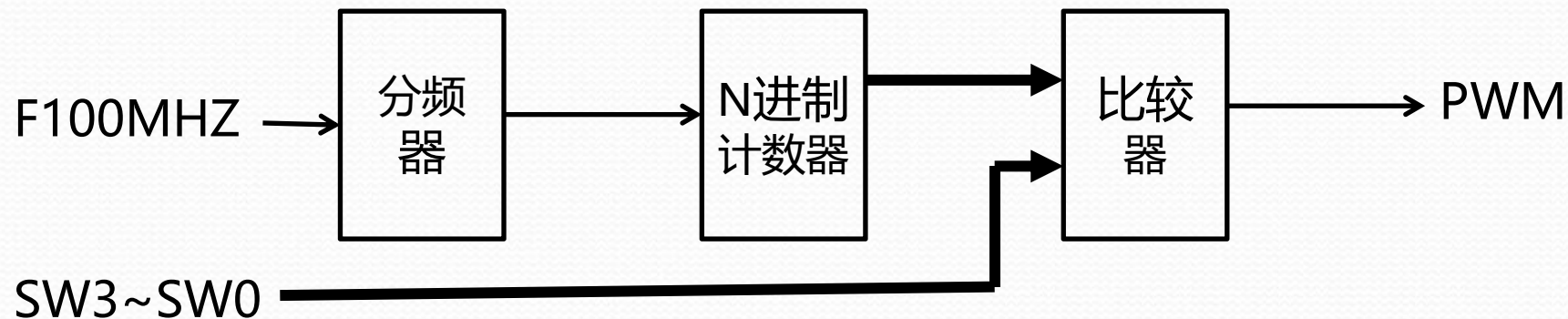
$$\delta = 0, \frac{1}{N} \times 100\%, \frac{2}{N} \times 100\% \dots$$

■ FPGA设计PWM的设计框图



- **分频器:** 将系统50MHz转换对应PWM要求的频率
- **N进制计数器:** 产生锯齿波(建议16档既可,16进制计数器)
- **比较器**(若计数器是16进制,输出是4位二进制,比较器 2个4位二进制比较)

■ PWM发生器需编写的VHDL源码



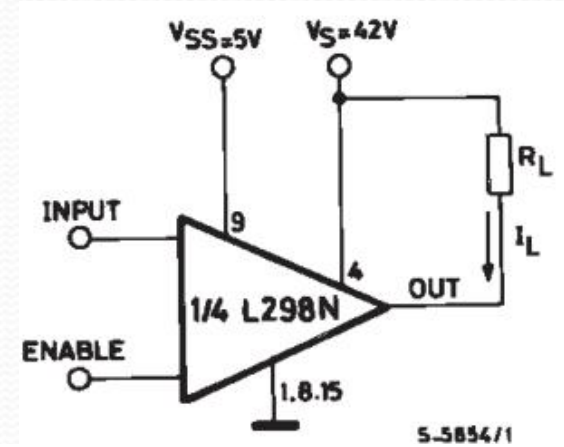
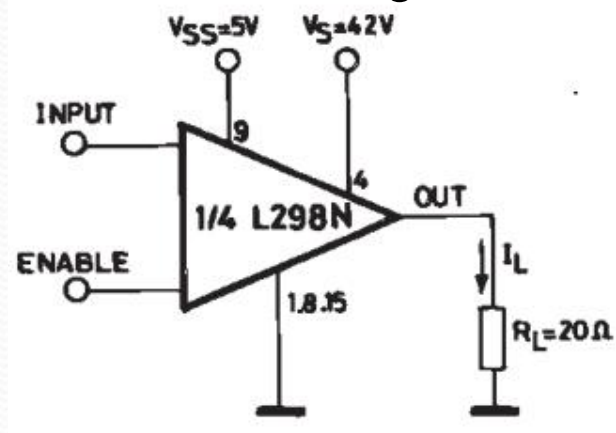
- 分频器VHDL源码
- N进制计数器源码
- 比较器源码
- 顶层源码

◆ 分频器参考源码



Figure3 : SourceCurrentDelay Times vs. Input orEnableSwitching.

Figure5 : SinkCurrent Delay Times vs. Input 0 V EnableSwitching.



ELECTRICAL CHARACTERISTICS (continued)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
T ₁ (V _i)	Source Current Turn-off Delay	0.5 V _i to 0.9 I _L (2); (4)		1.5		μs
T ₂ (V _i)	Source Current Fall Time	0.9 I _L to 0.1 I _L (2); (4)		0.2		μs
T ₃ (V _i)	Source Current Turn-on Delay	0.5 V _i to 0.1 I _L (2); (4)		2		μs
T ₄ (V _i)	Source Current Rise Time	0.1 I _L to 0.9 I _L (2); (4)		0.7		μs
T ₅ (V _i)	Sink Current Turn-off Delay	0.5 V _i to 0.9 I _L (3); (4)		0.7		μs
T ₆ (V _i)	Sink Current Fall Time	0.9 I _L to 0.1 I _L (3); (4)		0.25		μs
T ₇ (V _i)	Sink Current Turn-on Delay	0.5 V _i to 0.9 I _L (3); (4)		1.6		μs
T ₈ (V _i)	Sink Current Rise Time	0.1 I _L to 0.9 I _L (3); (4)		0.2		μs
fc (V _i)	Commutation Frequency	I _L = 2A		25	40	KHz

```

library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use IEEE.std_logic_unsigned.all;

entity DIVX is
    port (
        Freq_in,Clr_n : in  std_logic;
        Div_num:in natural range 0 to 49999999;
        Freq_out: out  std_logic);
end DIVX ;

architecture BEHAV of DIVX  is

begin
process(Freq_in,Clr_n)
variable cnt_val :natural range 0 to 49999999;
begin
    if(Clr_n='0') then
        cnt_val:=0;
    elsif(Freq_in'event and Freq_in='1') then
        if(cnt_val=Div_num) then
            cnt_val:=0;
        elsif(cnt_val<Div_num/2) then
            cnt_val:=cnt_val+1;
            Freq_out<='0';
        else
            cnt_val:=cnt_val+1;
            Freq_out<='1';
        end if;
    end if;

end process;

end BEHAV;
    
```


◆ 16进制计数器参考源码

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use IEEE.std_logic_unsigned.all;

entity CNT16 is
    port (
        Clk,Clr_n : in  std_logic;
        Q: out  std_logic_vector(3 downto 0));
end CNT16 ;

architecture BEHAV of CNT16  is
    signal Q_tmp: std_logic_vector(3 downto 0);
begin

    process (Clk,Clr_n)

    begin
        if(Clr_n='0') then
            Q_tmp<="0000";
        elsif(Clk'event and Clk='1') then
            Q_tmp<=Q_tmp+1;
        end if;

    end process;

    Q<=Q_tmp;

end BEHAV;
```


◆ 二个四位二进制比较器参考源码

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use IEEE.std_logic_unsigned.all;

entity COMP4BIT is
    port (
        A,B: in  std_logic_vector(3 downto 0);
        AGB:out  std_logic);
end COMP4BIT ;

architecture BEHAV of COMP4BIT  is

begin

    process (A,B)
    begin
        if (A>B) then
            AGB<='1';
        else
            AGB<='0';
        end if;

    end process;

end BEHAV;
```

□ 实验内容

- (1) 应用QUARTUS软件设计、仿真PWM电路
 - ①编写数字PWM产生电路的所有VHDL设计源码，并编译通过
 - ②编写数字PWM电路中比较器的VHDL仿真源码，并给出仿真波形
 - ③编写数字PWM电路中16进制计数器的VHDL仿真源码，并给出仿真波形
- (2) 基于FPGA开发板用QUARTUS软件综合、布线并产生并生成下载文件
- (3) 下载数字PWM产生电路，通过按键设定2种占空比，用示波器观察并记录PWM的输出波形，测量其周期、幅度和占空比。
- (4) 将PWM输出与电机驱动模块的总控输入ENA连接， $IN1=1, IN2=0$ ，观察不同占空比时电机的运动速度


```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use IEEE.std_logic_unsigned.all;
```

entity trace1 is

port

```
( LED6,LED7: in std_logic;
  R13,L13:out std_logic;
  nclr,clk_sys: in std_logic;
  PWM_in: in std_logic_vector(3 downto 0);
  PWM_out: out std_logic);
```

end trace1;

architecture BEHAV of trace1 is

Component CNT16 is

port (

```
  Clk,Clr_n: in std_logic;
  Q: out std_logic_vector(3 downto 0);
  CO:out std_logic);
```

end Component;

Component COMP4BIT is

port (

```
  clk: in std_logic;
  A,B: in std_logic_vector(3 downto 0);
  AGB:out std_logic);
```

end Component;

Component divX is

port (

```
  Freq_in,Clr_n : in std_logic;
  Div_num:in natural range 0 to 499999999;
  Freq_out: out std_logic);
```

end Component;


```

signal D: std_logic_vector(1 downto 0);
signal clk_temp,clk_comp: std_logic;
signal A_temp: std_logic_vector (3 downto 0);
begin
    U1:divx port
    map(Freq_in=>clk_sys,clr_n=>nclr,div_num=>
    2000,Freq_out=>clk_temp);
    U2:CNT16 port
    map(clk=>clk_temp,clr_n=>nclr,Q=>A_temp,C
    O=>clk_comp);
    U3:COMP4BIT port
    map(clk=>clk_comp,A=>A_temp,B=>PWM_in
    ,AGB=>PWM_out);
    D<= LED7 & LED6;

```

```

process(D)
    begin
        case D is
            when "00" => L13<='0';R13<='0';
            when "01" => L13<='1';R13<='0';
            when "10" => L13<='0';R13<='1';
            when "11" => L13<='1';R13<='1';
            when others => null;
        end case;
    end process;
end BEHAV;

```

任务总结：

1. 完成三个子功能模块调试和逻辑功能仿真；
2. 其中分频器模块通过三种不同的频率，测试并观察“欠调，正常，过调”

电机工作情况

（需自行设计测试方法）；

3. 用元件例化的方法编写PWM控制程序，在示波器上观察PWM波形。

（分别设定占空比**20%**，**50%**，**80%**）